

This result satisfies an important consistency check. Recall from Section 27.4 that any superpotential has dimensionality +3 (counting powers of mass, with $\hbar = c = 1$), while the scalar superfields Φ like ordinary scalar fields have dimensionality +1, so the Φ -dependent part of v_λ must appear with a coefficient of dimensionality

$$3 + \frac{2C_2}{C_1 - C_2} = \frac{3C_1 - C_2}{C_1 - C_2}.$$

This coefficient does not depend on the gauge coupling or on any of the couplings or masses in the superpotential, and because we have replaced the bare coupling g with g_λ in the second term of Eq. (29.3.11) it cannot depend on the ultraviolet cut-off K used to define g either, so it can only depend on λ . Therefore

$$v_\lambda(\Phi) = \lambda^{(3C_1 - C_2)/(C_1 - C_2)} H(\Phi), \quad (29.3.21)$$

where $H(\Phi)$ is a homogeneous function of order $-2n(d)C_{2d}/(C_1 - C_2)$ in the Φ s that belong to each representation d of the gauge group and is independent of any parameters of the theory. We can rewrite Eq. (29.3.14) in the form

$$\tau_\lambda = i \frac{3C_1 - C_2}{2\pi} \ln \left(\frac{\lambda}{\Lambda} \right) + \frac{\theta}{2\pi}, \quad (29.3.22)$$

where Λ is an energy parameter that characterizes the running gauge coupling, like the $\Lambda \approx 200$ MeV of quantum chromodynamics. Thus the last term in Eq. (29.3.15) is

$$\exp \left(\frac{2i\pi\tau_\lambda}{C_1 - C_2} \right) v_\lambda(\Phi) = \exp \left(\frac{i\theta}{C_1 - C_2} \right) \Lambda^{(3C_1 - C_2)/(C_1 - C_2)} H(\Phi). \quad (29.3.23)$$

The whole effective superpotential, including the non-perturbative contribution (29.3.23), is therefore independent of the floating cut-off λ .

The function $H(\Phi)$ is homogeneous and of negative order in Φ , so in the absence of a bare superpotential the potential is positive-definite at finite values of the scalar fields and vanishes only at infinite field values. In such a theory, there is no stable vacuum state, and the question of supersymmetry breaking is moot. The vacuum may be stabilized by adding a suitable bare superpotential. For instance, in generalized supersymmetric quantum chromodynamics with $N_f < N_c$ the only renormalizable bare superpotential is a sum of mass terms

$$f(Q, \bar{Q}) = \sum_{ij} m_{ij} \bar{Q}_{ai} Q_{aj}. \quad (29.3.24)$$

To seek a supersymmetric vacuum state, we need first of all to find what